

SECTION 4

8-BIT BUS OPERATION

The following paragraphs describe control signal and bus operation for 8-bit operation during data transfer operations, bus arbitration, bus error and halt conditions, and reset operation. The 8-bit bus operations devices are the MC68008, MC68HC001 in 8-bit mode, and MC68EC000 in 8-bit mode. The MC68HC001 and MC68EC000 select 8-bit mode by grounding mode during reset.

4.1 DATA TRANSFER OPERATIONS

Transfer of data between devices involves the following signals:

1. Address bus A0 through highest numbered address line
2. Data bus D0 through D7
3. Control signals

The address and data buses are separate parallel buses used to transfer data using an asynchronous bus structure. In all cases, the bus master must deskew all signals it issues at both the start and end of a bus cycle. In addition, the bus master must deskew the acknowledge and data signals from the slave device. For the MC68HC001 and MC68EC000, $\overline{UDS}$ is held negated and D15–D8 are undefined in 8-bit mode.

The following paragraphs describe the read, write, read-modify-write, and CPU space cycles. The indivisible read-modify-write cycle implements interlocked multiprocessor communications. A CPU space cycle is a special processor cycle.

4.1.1 Read Cycle

During a read cycle, the processor receives one byte of data from the memory or from a peripheral device. When the data is received, the processor internally positions the byte appropriately.

The 8-bit operation must perform two or four read cycles to access a word or long word, asserting the data strobe to read a single byte during each cycle. The address bus in 8-bit operation includes A0, which selects the appropriate byte for each read cycle. Figure 4-1 and 4-2 illustrate the byte read-cycle operation.

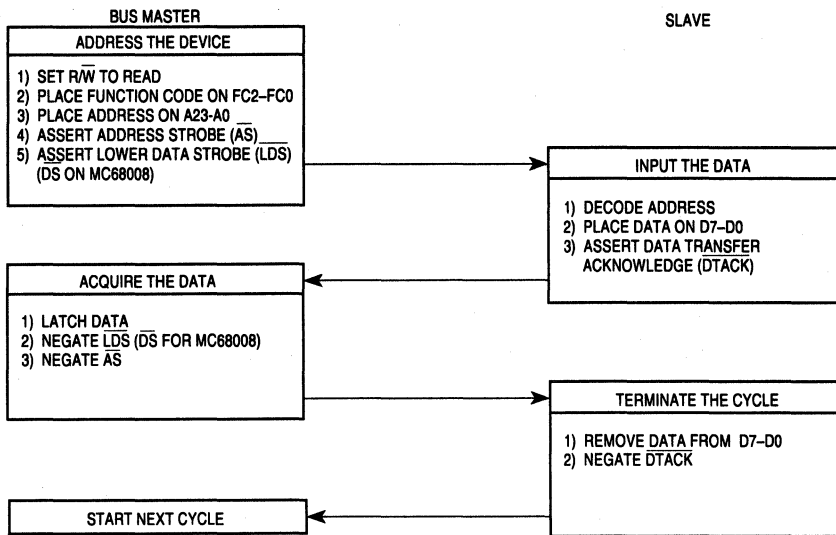


Figure 4-1. Byte Read-Cycle Flowchart

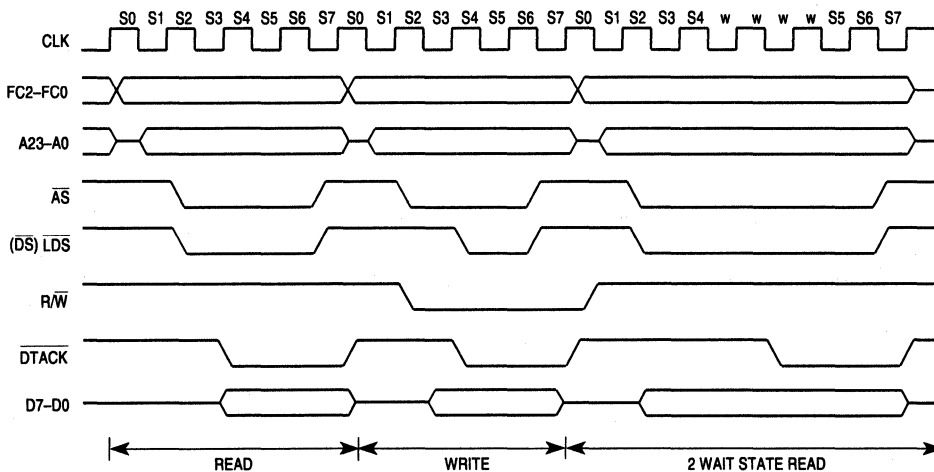


Figure 4-2. Read and Write-Cycle Timing Diagram

A bus cycle consists of eight states. The various signals are asserted during specific states of a read cycle, as follows:

- STATE 0 The read cycle starts in state 0 (S0). The processor places valid function codes on FC0–FC2 and drives $R/\overline{W}$ high to identify a read cycle.
- STATE 1 Entering state 1 (S1), the processor drives a valid address on the address bus.
- STATE 2 On the rising edge of state 2 (S2), the processor asserts $\overline{AS}$ and $\overline{LDS}$, or $\overline{DS}$.
- STATE 3 During state 3 (S3), no bus signals are altered.
- STATE 4 During state 4 (S4), the processor waits for a cycle termination signal ($\overline{DTACK}$ or $\overline{BERR}$) or $\overline{VPA}$, an M6800 peripheral signal. When $\overline{VPA}$ is asserted during S4, the cycle becomes a peripheral cycle (refer to **Appendix B M6800 Peripheral Interface**). If neither termination signal is asserted before the falling edge at the end of S4, the processor inserts wait states (full clock cycles) until either $\overline{DTACK}$ or $\overline{BERR}$ is asserted.
- STATE 5 During state 5 (S5), no bus signals are altered.
- STATE 6 During state 6 (S6), data from the device is driven onto the data bus.
- STATE 7 On the falling edge of the clock entering state 7 (S7), the processor latches data from the addressed device and negates $\overline{AS}$ and $\overline{LDS}$, or $\overline{DS}$. At the rising edge of S7, the processor places the address bus in the high-impedance state. The device negates $\overline{DTACK}$ or $\overline{BERR}$ at this time.

NOTE

During an active bus cycle, $\overline{VPA}$ and $\overline{BERR}$ are sampled on every falling edge of the clock beginning with S4, and data is latched on the falling edge of S6 during a read cycle. The bus cycle terminates in S7, except when $\overline{BERR}$ is asserted in the absence of $\overline{DTACK}$. In that case, the bus cycle terminates one clock cycle later in S9.

4.1.2 Write Cycle

During a write cycle, the processor sends bytes of data to the memory or peripheral device. Figures 4-3 and 4-4 illustrate the write-cycle operation

The 8-bit operation performs two write cycles for a word write operation, issuing the data strobe signal during each cycle. The address bus includes the A0 bit to select the desired byte.

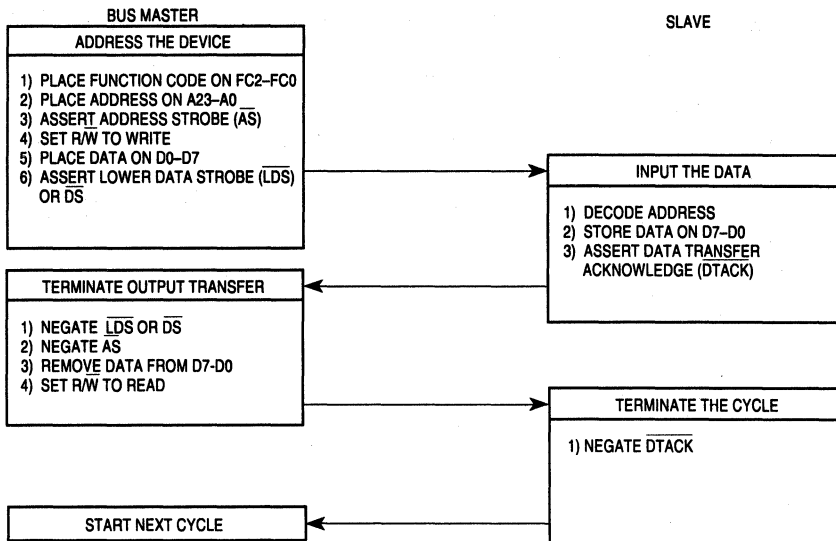


Figure 4-3. Byte Write-Cycle Flowchart

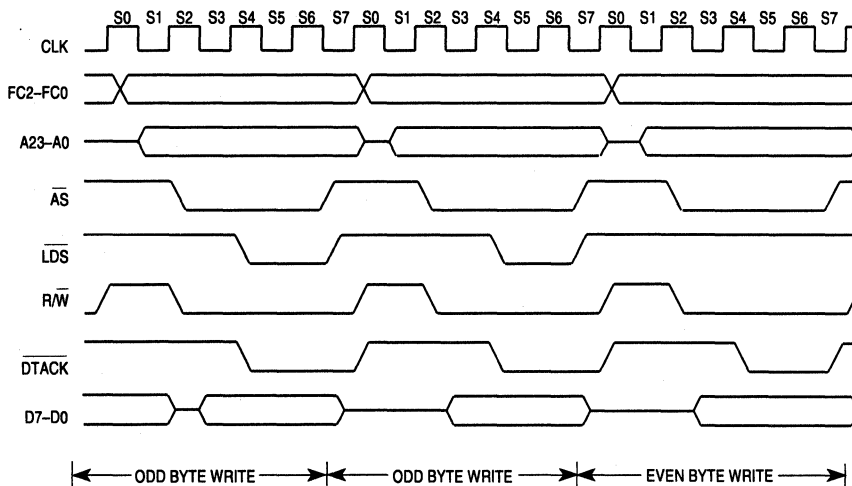


Figure 4-4. Write-Cycle Timing Diagram

The descriptions of the eight states of a write cycle are as follows:

- STATE 0 The write cycle starts in S0. The processor places valid function codes on FC2–FC0 and drives $R/\overline{W}$ high (if a preceding write cycle has left $R/\overline{W}$ low).
- STATE 1 Entering S1, the processor drives a valid address on the address bus.
- STATE 2 On the rising edge of S2, the processor asserts $\overline{AS}$ and drives $R/\overline{W}$ low.
- STATE 3 During S3, the data bus is driven out of the high-impedance state as the data to be written is placed on the bus.
- STATE 4 At the rising edge of S4, the processor asserts $\overline{LDS}$, or $\overline{DS}$. The processor waits for a cycle termination signal ($\overline{DTACK}$ or $\overline{BERR}$) or $\overline{VPA}$, an M6800 peripheral signal. When $\overline{VPA}$ is asserted during S4, the cycle becomes a peripheral cycle (refer to **Appendix B M6800 Peripheral Interface**). If neither termination signal is asserted before the falling edge at the end of S4, the processor inserts wait states (full clock cycles) until either $\overline{DTACK}$ or $\overline{BERR}$ is asserted.
- STATE 5 During S5, no bus signals are altered.
- STATE 6 During S6, no bus signals are altered.
- STATE 7 On the falling edge of the clock entering S7, the processor negates $\overline{AS}$, $\overline{LDS}$, and $\overline{DS}$. As the clock rises at the end of S7, the processor places the address and data buses in the high-impedance state, and drives $R/\overline{W}$ high. The device negates $\overline{DTACK}$ or $\overline{BERR}$ at this time.

4.1.3 Read-Modify-Write Cycle.

The read-modify-write cycle performs a read operation, modifies the data in the arithmetic logic unit, and writes the data back to the same address. The address strobe ($\overline{AS}$) remains asserted throughout the entire cycle, making the cycle indivisible. The test and set (TAS) instruction uses this cycle to provide a signaling capability without deadlock between processors in a multiprocessing environment. The TAS instruction (the only instruction that uses the read-modify-write cycle) only operates on bytes. Thus, all read-modify-write cycles are byte operations. Figure 4-5 and 4-6 illustrate the read-modify-write cycle operation.

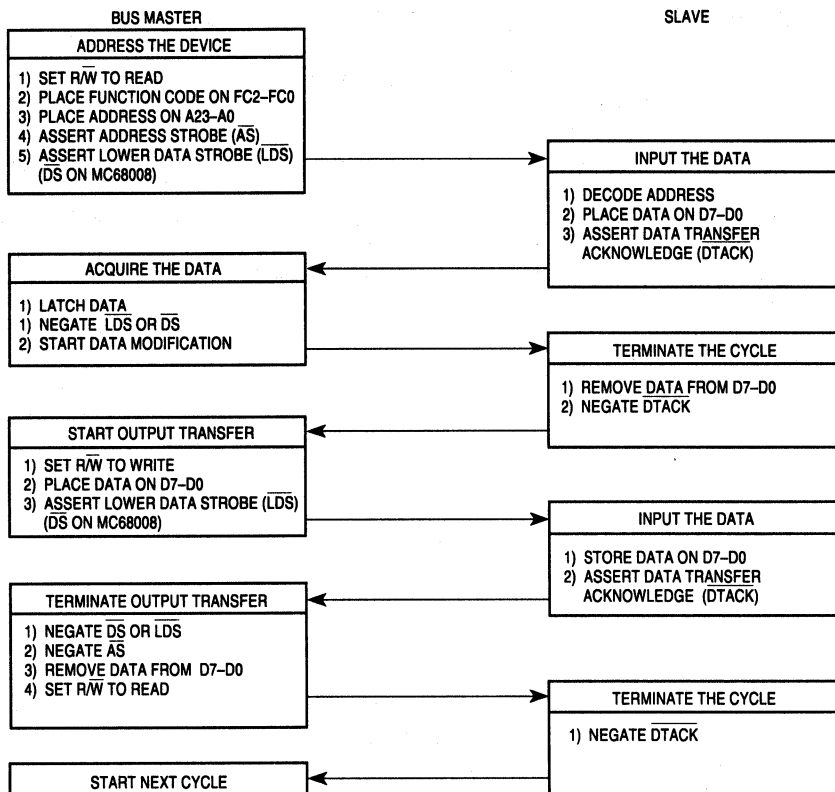


Figure 4-5. Read-Modify-Write Cycle Flowchart

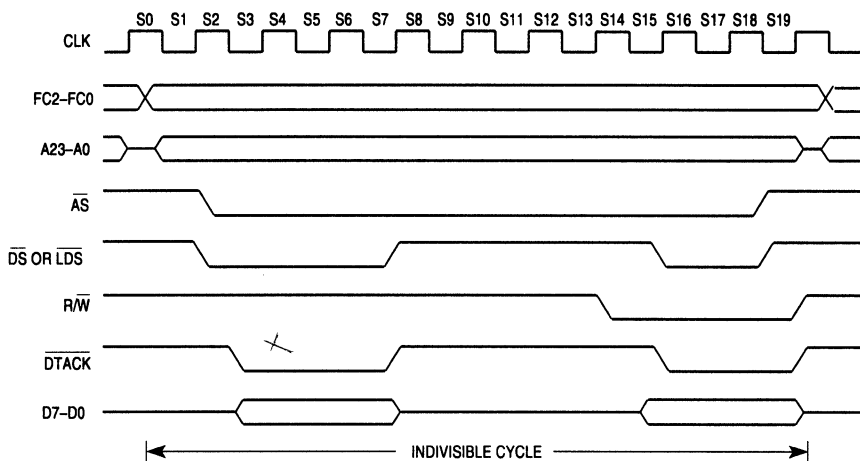


Figure 4-6. Read-Modify-Write Cycle Timing Diagram

The descriptions of the read-modify-write cycle states are as follows:

- STATE 0** The read cycle starts in S0. The processor places valid function codes on FC2-FC0 and drives R/W high to identify a read cycle.
- STATE 1** Entering S1, the processor drives a valid address on the address bus.
- STATE 2** On the rising edge of S2, the processor asserts $\overline{AS}$ and $\overline{LDS}$, or $\overline{DS}$.
- STATE 3** During S3, no bus signals are altered.
- STATE 4** During S4, the processor waits for a cycle termination signal ($\overline{DTACK}$ or $\overline{BERR}$) or $\overline{VPA}$, an M6800 peripheral signal. When $\overline{VPA}$ is asserted during S4, the cycle becomes a peripheral cycle (refer to **Appendix B M6800 Peripheral Interface**). If neither termination signal is asserted before the falling edge at the end of S4, the processor inserts wait states (full clock cycles) until either $\overline{DTACK}$ or $\overline{BERR}$ is asserted.
- STATE 5** During S5, no bus signals are altered.
- STATE 6** During S6, data from the device are driven onto the data bus.
- STATE 7** On the falling edge of the clock entering S7, the processor accepts data from the device and negates $\overline{LDS}$, and $\overline{DS}$. The device negates $\overline{DTACK}$ or $\overline{BERR}$ at this time.
- STATES 8-11** The bus signals are unaltered during S8-S11, during which the arithmetic logic unit makes appropriate modifications to the data.

- STATE 12 The write portion of the cycle starts in S12. The valid function codes on FC2–FC0, the address bus lines, $\overline{AS}$, and $R/\overline{W}$ remain unaltered.
- STATE 13 During S13, no bus signals are altered.
- STATE 14 On the rising edge of S14, the processor drives $R/\overline{W}$ low.
- STATE 15 During S15, the data bus is driven out of the high-impedance state as the data to be written are placed on the bus.
- STATE 16 At the rising edge of S16, the processor asserts $\overline{LDS}$ or $\overline{DS}$. The processor waits for $\overline{DTACK}$ or $\overline{BERR}$ or $\overline{VPA}$, an M6800 peripheral signal. When $\overline{VPA}$ is asserted during S16, the cycle becomes a peripheral cycle (refer to **Appendix B M6800 Peripheral Interface**). If neither termination signal is asserted before the falling edge at the close of S16, the processor inserts wait states (full clock cycles) until either $\overline{DTACK}$ or $\overline{BERR}$ is asserted.
- STATE 17 During S17, no bus signals are altered.
- STATE 18 During S18, no bus signals are altered.
- STATE 19 On the falling edge of the clock entering S19, the processor negates $\overline{AS}$, $\overline{LDS}$, and $\overline{DS}$. As the clock rises at the end of S19, the processor places the address and data buses in the high-impedance state, and drives $R/\overline{W}$ high. The device negates $\overline{DTACK}$ or $\overline{BERR}$ at this time.

4.2 OTHER BUS OPERATIONS

Refer to **Section 5 16-Bit Bus Operations** for information on the following items:

- CPU Space Cycle
- Bus Arbitration
 - Bus Request
 - Bus Grant
 - Bus Acknowledgment
- Bus Control
- Bus Errors and Halt Operations
- Reset Operations
- Asynchronous Operations
- Synchronous Operations